| | | |
|---|---|---|
| \|\| | **OR** | The logical OR combines two different relational expressions in to one. It returns 1 (True), if either one of the expression is true. It returns 0 (false), if both the expressions are false. |
| **!** | **NOT** | NOT works on a single expression / operand. It simply negates or inverts the truth value. i.e., if an operand / expression is 1 (true) then this operator returns 0 (false) and vice versa |

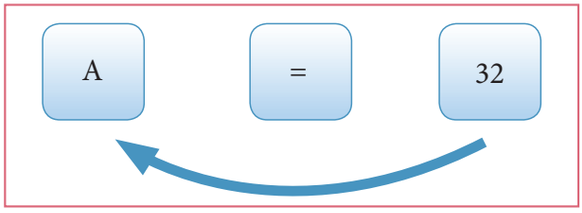
- AND, OR both are binary operators where as NOT is an unary operator.

**Example:** a = 5, b = 6, c = 7;

| Expression | Result |
|---|---|
| (a<b) && (b<c) | 1 (True) |
| (a>b) && (b<c) | 0 (False) |
| (a<b) \|\| (b>c) | 1 (True) |
| !(a>b) | 1 (True) |

## (4)Assignment Operator:

Assignment operator is used to assign a value to a variable which is on the left hand side of an assignment statement. = (equal to) is commonly used as the assignment operator in all computer programming languages. This operator copies the value at the right side of the operator to the left side variable. It is also a binary operator.




C++ uses different types of assignment operators. They are called as Shorthand assignment operators.

| Operator | Name of Operator | Example |
|---|---|---|
| += | Addition Assignment | a = 10;<br>c = a += 5;<br>(ie, a = a + 5)<br>c = 15 |
| -= | Subtraction Assignment | a = 10;<br>c = a -= 5;<br>(ie. a = a – 5)<br>c = 5 |
| *= | Multiplication Assignment | a = 10;<br>c = a *= 5;<br>(ie. a = a * 5)<br>c = 50 |
| /= | Division Assignment | a = 10;<br>c = a /= 5;<br>(ie. a = a / 5)<br>c = 2 |
| %= | Modulus Assignment | a = 10;<br>c = a %= 5;<br>(ie. a = a % 5)<br>c = 0 |

Discuss the differences between = and == operators

## (5) Conditional Operator:

In C++, there is only one conditional operator. **?:** is a conditional Operator which is also known as Ternary Operator. This operator is used as an alternate to if … else control statement. We will learn more about this operator in later chapters along with if …. else structure.

## Other Operators:

| | |
|---|---|
| The Comma operator | Comma ( , ) is an operator in C++ used to string together several expressions. The group of expression separated by comma is evaluated from left to right. |
| Sizeof | This is called as compile time operator. It returns the size of a variable in bytes. |
| Pointer | * Pointer to a variable<br>& Address of |



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